CertGuard AI

SSL Certificate Watcher & AI Agent

Scan · Classify · Recommend · Monitor

Overview

CertGuard AI is a proactive SSL certificate monitoring system that automatically scans registered domains, retrieves certificate metadata, calculates remaining validity, classifies risk, and generates AI-powered remediation recommendations via Groq API (Llama 3). The platform also delivers real-time dashboard analytics, historical scan tracking, email alerting, and on-demand report exports — enabling organisations to manage certificate lifecycles before outages occur.

Problem Statement

Challenge	Impact
No centralised certificate monitoring	Missed expiry across multiple domains
Manual expiry tracking	Human error and operational overhead
Unexpected certificate expiration	Browser warnings, service disruptions
Reactive rather than proactive management	Reputational and security risk

Objectives

- Automatically scan SSL certificates for all monitored domains.
- Retrieve issuer, validity dates, and serial number via TLS handshake.
- Calculate days remaining and classify risk: Critical (≤7 d), Warning (≤30 d), Healthy (>30 d).
- Generate AI-powered renewal tickets (title, priority, business impact, action steps) via Groq/Llama 3.
- Store all scan results and history in SQLite for trend analysis and audit.
- Deliver email alerts via Gmail SMTP for Critical and Warning certificates.
- Export reports in CSV and Markdown formats on demand.
- Provide a React dashboard with real-time summary cards and charts.

System Actors

Actor	Role
User / Admin	Adds domains, triggers scans, views dashboard, exports reports.
SSL Scanner	Connects to port 443, performs TLS handshake, extracts metadata.
AI Agent (Groq/Llama 3)	Analyses risk; generates ticket title, priority & remediation steps.
SQLite Database	Persists domains, scan results, risk tags, AI tickets, audit history.
Email Alert System	Sends SMTP notifications for Critical and Warning certificates.
Export Engine	Produces CSV and Markdown reports on demand.

Use Case Summary

UC#	Use Case	Actor	Description
UC1	Manage Domains	User	Add, validate, and list domains for monitoring.
UC2	Scan SSL Certificate	SSL Scanner	Port 443 TLS handshake; extract issuer, dates, serial.
UC3	Classify Risk Level	Rule Engine	Tag cert: Critical (≤7 d), Warning (≤30 d), Healthy (>30 d).
UC4	Generate AI Recommendation	AI Agent	Groq/Llama 3 produces ticket: title, priority, action.
UC5	View Dashboard	User	Real-time summary cards, doughnut & bar charts.
UC6	Track Scan History	SQLite	Append-only scan log for trend analysis and audit.
UC7	Send Email Alert	Alert System	Gmail SMTP notification on Critical/Warning transition.
UC8	Export Report	Export Engine	CSV or Markdown report of current scan results.

Technology Stack

Layer	Technology
Frontend	React, TypeScript, HTML5, CSS3
Backend	Python, FastAPI, Socket Programming, SSL Libraries
Database	SQLite
AI / LLM	Groq API · Llama 3 (llama3-8b-8192)
Alerts	Gmail SMTP via smtplib
Reporting	CSV Export, Markdown Export, pandas

System Workflow

Step	Action
1	User enters a domain name in the React dashboard.
2	SSL Scanner connects to domain:443 and performs a TLS handshake.
3	Certificate metadata (issuer, dates, serial) is extracted; days remaining calculated.
4	Risk Classification engine tags the certificate — Critical, Warning, or Healthy.
5	AI Agent (Groq/Llama 3) generates a structured remediation ticket for at-risk certs.
6	All results (domain, scan data, risk tag, AI ticket) are persisted to SQLite.
7	Dashboard analytics are refreshed with updated counts and visualisations.
8	Email Alert System notifies stakeholders if the certificate is Critical or Warning.
9	User exports a CSV or Markdown report for operational or documentation use.

Risk Classification Model

Days Remaining	Status	Required Action
7 days or fewer	**Critical**	Immediate renewal — page the on-call team.
8 – 30 days	**Warning**	Schedule renewal — send alert notification.
More than 30 days	**Healthy**	No action required — continue monitoring.

Key Features

Feature	Description
SSL Certificate Monitoring	Continuous automated scanning of all registered domains.
Expiry Detection	Precise calculation of days remaining per certificate.
Risk Classification	Three-tier severity model: Critical, Warning, Healthy.
AI-Powered Recommendations	Groq + Llama 3 generate actionable remediation tickets.
Dashboard Analytics	Real-time summary cards, doughnut and bar charts.
Historical Tracking	Append-only scan history in SQLite for trend analysis.
Email Alert Notifications	Gmail SMTP alerts on Critical / Warning state transitions.
CSV & Markdown Export	Tabular data export and formatted documentation reports.
SQLite Storage	Lightweight, zero-config persistent data layer.

Risks & Mitigations

Risk	Mitigation
Groq API unavailability	Fallback to rule-based summary; retry with exponential backoff.
SMTP delivery failure	Retry up to 3 times; log failure; surface in-app notification.
Domain unreachable on port 443	Log connection error; mark domain unreachable; alert user.
SQLite data loss	Scheduled DB file backup; export before destructive operations.
False risk classification	Hard rule overrides LLM output for ≤7 and ≤30 day thresholds.

Future Enhancements

- Slack and Microsoft Teams alert integrations.
- Automatic certificate renewal via ACME / Let's Encrypt.
- Multi-user authentication and role-based access control (RBAC).
- Cloud deployment on AWS, GCP, or Azure.
- Real-time continuous monitoring with configurable scan intervals.
- REST API for third-party and CI/CD pipeline integrations.

Success Metrics

Metric	Target
Zero unexpected expirations	All domains scanned at least once per day.
AI recommendation accuracy	≥ 95 % — Llama 3 tickets rated useful by admin.
Alert delivery rate	100 % — every Critical/Warning transition triggers email.
Scan time per domain	≤ 2 seconds for TLS handshake and metadata extraction.
Scan result persistence	100 % — no data loss between scan and SQLite write.

Conclusion

CertGuard AI delivers a robust, end-to-end solution for SSL certificate lifecycle management. By combining automated domain scanning, deterministic risk classification, LLM-driven remediation guidance, and proactive alerting, the system ensures no certificate expiry goes unnoticed. The lightweight SQLite backend and React-based dashboard make it straightforward to deploy and operate, while the export and alert capabilities integrate naturally into existing DevOps and IT workflows. With a clear roadmap for cloud deployment and automated renewal, CertGuard AI is designed to scale alongside organisational growth.

CertGuard AI · v1.0 · Proactive SSL Monitoring & AI-Powered Remediation